Lectures 22b-23

Plane Waves in Materials

- Perfect dielectric, $\sigma = 0$
- Imperfect dielectric,
- Good conductor,
- Perfect conductor $\sigma \to \infty$

Dr. Victoria Shao
yangshao@Illinois.edu

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Almost the entire Greek Alphabet!!!

σ

ε

μ

These constants are material properties and affect wave propagation: $\alpha, \beta, \gamma, \tau, \eta$

α

β

$\overline{\gamma}$

τ

$\overline{\eta}$

α Alpha	β Beta	γ Gamma	δ Delta	ε Epsilon	ζ Zeta
η Eta	θ Theta	ι Iota	κ Kappa	λ Lambda	μ Mu
ν Nu	ξ Xi	ο Omicron	π Pi	ρ Rho	σ Sigma
τ Tau	υ Upsilon	ϕ Phi	χ Chi	ψ Psi	ω Omega

Wave Equation for E_x

$$\frac{\partial^2 \tilde{E}_x}{\partial z^2} = \underbrace{j\omega\mu(\sigma + j\omega\epsilon)}\tilde{E}_x$$

$(\text{Propagation Constant } \bar{\gamma})^2$

One of the most important definition from last lecture

$$\frac{\partial^2 \tilde{E}_x}{\partial z^2} = \bar{\gamma}^2 \tilde{E}_x$$

For wave in free space, wave eqn was: $\frac{\partial^2 E_x}{\partial z^2} = \mu_0 \epsilon_0 \frac{\partial^2 E_x}{\partial t^2}$

Propagation Constant

$$\bar{\gamma} = \alpha + j\beta = \sqrt{j\omega\mu(\sigma + j\omega\varepsilon)}$$

Calculating $\bar{\gamma}^2$ gives 2 equations and 2 unknowns α, β

$$\bar{\gamma}^2 = \alpha^2 - \beta^2 + 2j\alpha\beta = -\omega^2\mu\varepsilon + j\omega\mu\sigma$$

$$\therefore\ \alpha^2 - \beta^2 = -\omega^2\mu\varepsilon \ \text{ and } \ 2\alpha\beta = \omega\mu\sigma$$

Solving for α and β using Mathematica:

$$\alpha = \omega\sqrt{\frac{\mu\varepsilon}{2}}\left(\sqrt{1+\left(\frac{\sigma}{\omega\varepsilon}\right)^2} - 1\right)^{1/2}$$

Attenuation Constant

$$\beta = \omega\sqrt{\frac{\mu\varepsilon}{2}}\left(\sqrt{1+\left(\frac{\sigma}{\omega\varepsilon}\right)^2} + 1\right)^{1/2}$$

Phase Constant

The key relationships

$$\alpha = \omega\sqrt{\frac{\mu\varepsilon}{2}}\left(\sqrt{1+\left(\frac{\sigma}{\omega\varepsilon}\right)^2}-1\right)^{1/2}$$ attenuation

$$e^{-\alpha z}$$

$$\beta = \omega\sqrt{\frac{\mu\varepsilon}{2}}\left(\sqrt{1+\left(\frac{\sigma}{\omega\varepsilon}\right)^2}+1\right)^{1/2}$$ velocity

$$v_p = \omega/\beta$$

Wavelength

$$\lambda = \frac{2\pi}{\beta}$$

Don't need to memorize these equations!

Attenuation and velocity are <u>functions</u> of ω (dispersion)

Complex Material Impedance

$$\overline{\eta} = \sqrt{\frac{j\omega\mu}{\sigma + j\omega\varepsilon}}$$

$$\overline{\gamma} = \sqrt{(j\omega\mu)(\sigma + j\omega\varepsilon)}$$

$$\overline{\gamma}\,\overline{\eta} = j\omega\mu$$

$$\overline{\eta} = |\overline{\eta}|e^{j\tau}$$

Magnitude of Complex Impedance

Phase difference between E_x and H_y

Next: Solving for $|\eta|$ and τ

Solving for |η| and τ

We know $\bar{\gamma} \equiv \sqrt{j\omega\mu(\sigma + j\omega\varepsilon)}$

$$\bar{\eta} \equiv \sqrt{\frac{j\omega\mu}{\sigma + j\omega\varepsilon}}$$

$$\therefore \tilde{E} = \pm\bar{\eta}\tilde{H} \text{ where } \bar{\eta} = \frac{j\omega\mu}{\bar{\gamma}}$$

$$\bar{\eta}^2 = |\bar{\eta}|^2 e^{j2\tau} = \frac{j\omega\mu(\sigma - j\omega\varepsilon)}{\sigma^2 + (\omega\varepsilon)^2} = \frac{\omega\mu(\omega\varepsilon + j\sigma)}{\sigma^2 + (\omega\varepsilon)^2}$$

$$|\eta| = \sqrt{\frac{\omega\mu}{\sqrt{\sigma^2 + (\omega\varepsilon)^2}}}$$

$$\therefore \tau = \frac{1}{2}\tan^{-1}\left(\frac{\sigma}{\omega\varepsilon}\right)$$

Don't need to memorize these two equations!

Propagation in Material

For a uniform plane wave with $f = 10^6$ Hz in a nonmagnetic medium ($\mu=\mu_0$), the propagation constant is: $\bar{\gamma} = 0.05 + 0.1j\ m^{-1}$. Find:

σ. ε.

α β

skin depth. δ.

a) Distance for field to attenuate by e^{-1}

$e^{-\alpha z} = e^{-1} \Rightarrow \alpha z = 1 \Rightarrow z = \frac{1}{\alpha} = \frac{1}{0.05} = 20\,[m]$

b) Distance for field to change phase by 1 rad

$\cos(\omega t \mp \beta z)$ $\quad \beta z = 1 \quad z = \frac{1}{\beta} = \frac{1}{0.1} = 10\,[m]$

c) Distance that constant phase moves in $1\mu s$

$dis = v_p t = \frac{\omega}{\beta} t = \frac{2\pi\times10^6}{0.1}\times10^{-6} = 20\pi\ [m]$

d) The ratio of |**E**| to |**H**|

$|\eta| = \frac{|E|}{|H|}$

$\bar{\gamma}\bar{\eta} = j\omega\mu \Rightarrow \bar{\eta} = \frac{j\omega\mu}{\bar{\gamma}} = \frac{j\cdot2\pi\times10^6\times4\pi\times10^{-7}}{0.05+0.1j}$

$|\eta|e^{j\tau}$ $\quad |\bar{\eta}|$.

e) The phase difference between **E** and **H**

$\angle E = \angle H + \tau$. $\quad \tau = \frac{\pi}{2} - \tan^{-1}\left(\frac{0.1}{0.05}\right)$

Plane Waves in Materials

Four special cases — α, β $|\eta|$ τ.

- Perfect dielectric,
- Imperfect dielectric, → v_p → λ
- Good conductor,
- Perfect conductor

Perfect Dielectric Material

Definition of a PERFECT dielectric: $\sigma = 0$

Propagation Constant:

$$\bar{\gamma} = \sqrt{j\omega\mu(\sigma + j\omega\varepsilon)}$$

$$\bar{\gamma} = j\omega\sqrt{\mu\varepsilon} = \alpha + j\beta$$

$\alpha = 0$.

Perfect Dielectric Material

$$\bar{\gamma} = j\omega\sqrt{\mu\varepsilon}$$

$$\bar{\gamma} = \alpha + j\beta$$

$$\alpha = 0$$

NO ATTENUATION

$$\beta = \omega\sqrt{\mu\varepsilon}$$

$$v_p = \frac{\omega}{\beta} = \frac{1}{\sqrt{\mu\varepsilon}}$$

Speed of wave is less than in free space:

$$v_p = \frac{1}{\sqrt{\mu\varepsilon}} = \frac{1}{\sqrt{\mu_0\mu_r\varepsilon_0\varepsilon_r}} = \frac{c}{\sqrt{\mu_r\varepsilon_r}}$$

$c = \frac{1}{\sqrt{\mu_0\varepsilon_0}}$.

$\lambda = \frac{2\pi}{\beta}$.

$= \frac{2\pi}{\omega\sqrt{\mu\varepsilon}}$.

$= \frac{2\pi}{\omega\sqrt{\mu_0\varepsilon_0}\sqrt{\mu_r\varepsilon_r}}$

$= \lambda_0 / \sqrt{\mu_r\varepsilon_r}$.

Perfect Dielectric Material

$$\overline{\eta} = \sqrt{\frac{j\omega\mu}{\sigma + j\omega\varepsilon}} = \sqrt{\frac{\mu}{\varepsilon}} = |\overline{\eta}|e^{j\tau}$$

($\sigma \to 0$)

Is a REAL number

$$\tau = 0$$

E and **H** are IN PHASE

$$\beta = \beta_0\sqrt{\mu_r\varepsilon_r}$$

$$v_p = \frac{c}{\sqrt{\mu_r\varepsilon_r}}$$

$$\lambda = \frac{\lambda_0}{\sqrt{\mu_r\varepsilon_r}}$$

$$\eta = \eta_0\sqrt{\frac{\mu_r}{\varepsilon_r}}$$

$$\frac{|E|}{|H|} = |\overline{\eta}| = \sqrt{\frac{\mu}{\varepsilon}} = \sqrt{\frac{\mu_0\mu_r}{\varepsilon_0\varepsilon_r}} = \eta_0\sqrt{\frac{\mu_r}{\varepsilon_r}}$$

$$\eta_0 = \sqrt{\frac{\mu_0}{\varepsilon_0}} = 120\pi\ [\Omega]$$

Impedance is different than free space - could be greater or less, depending on material

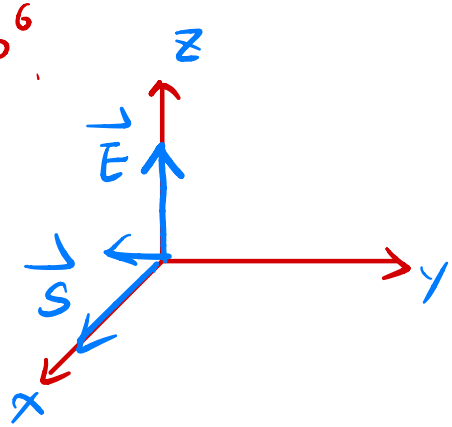
Finding parameters for perfect dielectric

$\vec{E} = 10\cos(\omega t - \beta x)\hat{z}$

Given $\mu=\mu_0$ and $\vec{E} = 10\cos(3\pi 10^7 t - 0.2\pi x)\hat{a}_z$, find:

(a) The frequency f

$\omega = 2\pi f = 3\pi\times 10^7 \Rightarrow f = 1.5\times 10^7 = 15\ MHz$ (10^6)



(b) The wavelength

$\lambda = \frac{2\pi}{\beta} = \frac{2\pi}{0.2\pi} = 10\ [m]$

(c) The phase velocity

$v_p = \frac{\omega}{\beta} = \frac{3\pi\times 10^7}{0.2\pi} = 1.5\times 10^8\ m/s = \frac{c}{2}$

(d) The relative permittivity ε_r ?

$v_p = \frac{c}{\sqrt{\mu_r\varepsilon_r}}$ $\mu_r = 1$ $\sqrt{\varepsilon_r} = \frac{c}{v_p} = \frac{c}{c/2} = 2 \Rightarrow \varepsilon_r = 4$

(e) The associated $\vec{H}$ field

$\vec{H} = \frac{10}{|\eta|}\cos(\omega t - \beta x - 0)(-\hat{y})$

$\eta = \eta_0\sqrt{\frac{\mu_r}{\varepsilon_r}} = \eta_0\sqrt{\frac{1}{4}} = \frac{\eta_0}{2}$

Imperfect Dielectric

Definition: $\sigma \neq 0$

But material does not conduct enough to really be considered a conductor

So, how much conductivity is "enough"?

Look at the **loss tangent:**

$$\frac{\sigma}{\omega\varepsilon}$$

σ → conductivity

$\omega\varepsilon$ → dielectric

Imperfect Dielectric

$$\frac{\sigma}{\omega\varepsilon} << 1$$

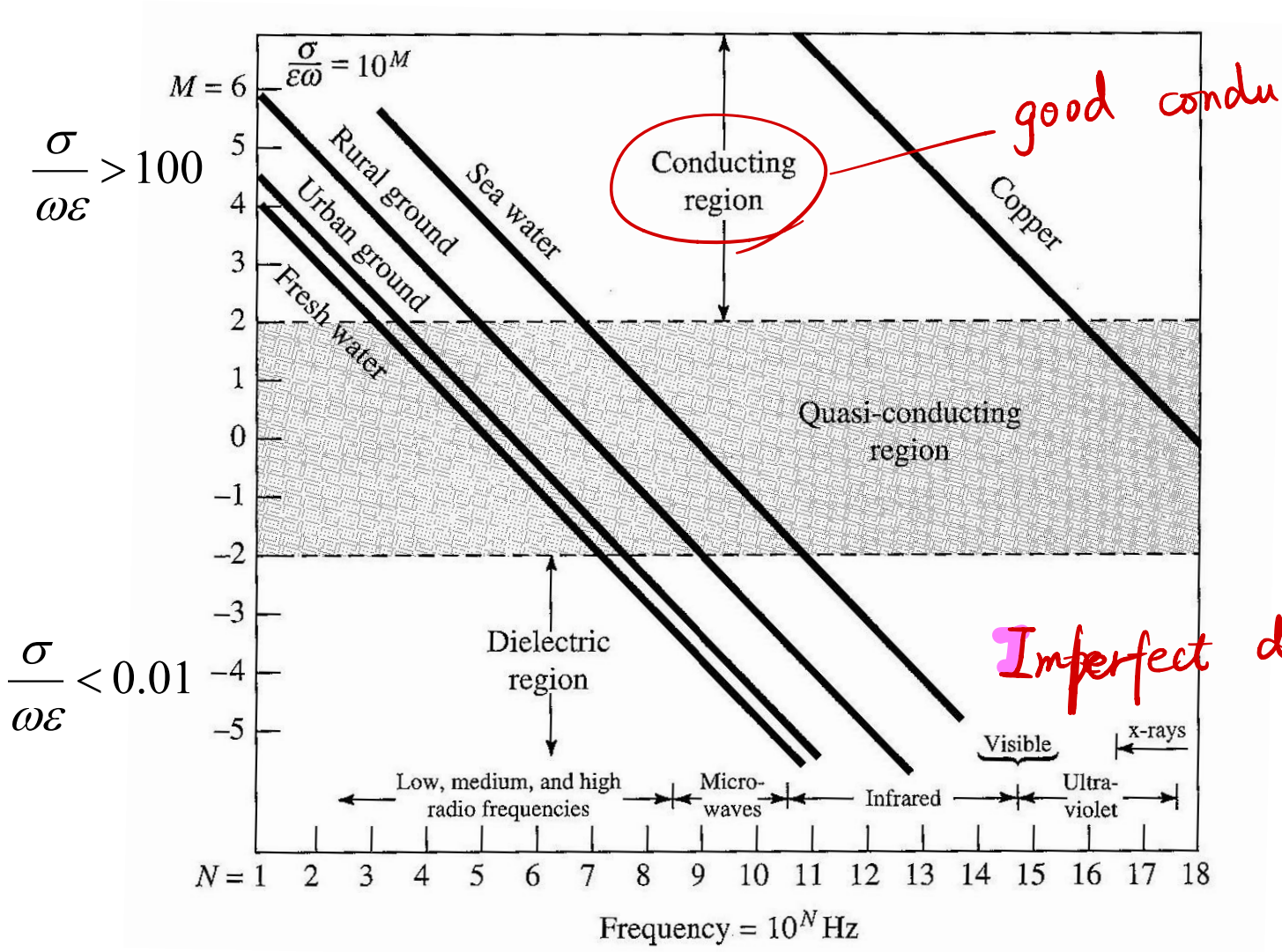
Official math definition for what constitutes an imperfect dielectric (small loss tangent)

Very important definition

$$\frac{\partial^2 \tilde{E}_x}{\partial z^2} = j\omega\mu(\sigma + j\omega\varepsilon)\tilde{E}_x$$

Tells which term is dominant

Behaves like a Dielectric or Conductor depending on ω



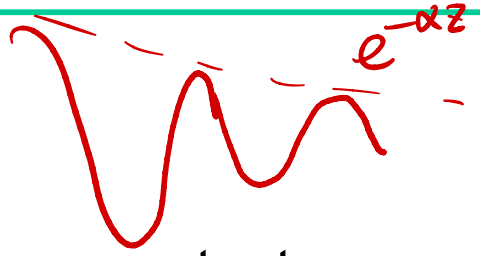
Imperfect Dielectric: How is it different from perfect dielectric?

Only important difference: There IS attenuation in an imperfect dielectric

$$\alpha \neq 0$$

$$\bar{\gamma} = \sqrt{j\omega\mu(\sigma + j\omega\varepsilon)}$$

$$\alpha \approx \frac{\sigma}{2}\sqrt{\frac{\mu}{\varepsilon}}$$ attenuation constant



Everything else is the same as in a perfect dielectric:

$$\beta \approx \omega\sqrt{\mu\varepsilon} \qquad v_p = \frac{\omega}{\beta} = \frac{1}{\sqrt{\mu\varepsilon}} \qquad \lambda = 2\pi/\beta$$

Pretty good approximation as long as $\frac{\sigma}{\omega\varepsilon} < 0.1$

Imperfect Dielectric

$$\frac{\sigma}{\omega\varepsilon} << 1$$

Complex Impedance

$$\bar{\eta} = \sqrt{\frac{j\omega\mu}{\sigma + j\omega\epsilon}} = |\bar{\eta}|e^{j\tau}$$

$$\bar{\eta} = \sqrt{\frac{\mu}{\epsilon\left(1 - j\frac{\sigma}{\omega\epsilon}\right)}}$$

$$\approx \sqrt{\frac{\mu}{\epsilon}}\left(1 + j\frac{\sigma}{2\omega\epsilon}\right)$$

$$\approx \sqrt{\frac{\mu}{\epsilon}} e^{j\tan^{-1}\frac{\sigma}{2\omega\epsilon}}$$

$$|\bar{\eta}| \approx \sqrt{\frac{\mu}{\epsilon}} = \eta_0\sqrt{\frac{\mu_r}{\epsilon_r}}$$

same as dielectric

$$\tau \approx \frac{\sigma}{2\omega\epsilon} \to 0$$

E and **H** are approximately IN PHASE

Taylor expansion

$$(1-x)^{-\frac{1}{2}} \approx 1 + \frac{1}{2}x + \cdots$$

$$\tan^{-1} x = x - \frac{x^3}{3} + \cdots$$

Good Conductors

Definition: $\sigma \neq \infty$

And conductivity is large

$$\frac{\sigma}{\omega\varepsilon} >> 1$$

Official math definition of
a good conductor
Large loss tangent

Note that any material with $\sigma>0$ behaves like a good conductor at low enough frequency

Good Conductor

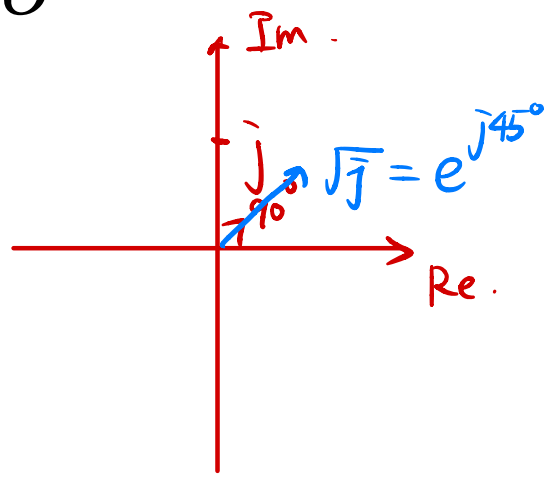
Propagation Constant

$$\bar{\gamma} = \sqrt{j\omega\mu(\sigma + j\omega\varepsilon)} \approx \sqrt{j\omega\mu\sigma}$$

$$= \alpha + j\beta$$

$$= \sqrt{j}\sqrt{\omega\mu\sigma}$$

Warning: math trick $\sqrt{j} = \dfrac{1+j}{\sqrt{2}}$ How?



$$\bar{\gamma} = \sqrt{\frac{\omega\mu\sigma}{2}}(1+j) = \alpha + j\beta$$

$$\boxed{\alpha = \beta = \sqrt{\frac{\omega\mu\sigma}{2}}}$$

Good Conductor

Complex Impedance

$$\overline{\eta} = \sqrt{\frac{j\omega\mu}{\sigma + j\omega\varepsilon}} \approx \sqrt{\frac{j\omega\mu}{\sigma}} = \sqrt{j}\sqrt{\frac{\omega\mu}{\sigma}}$$

$\sigma \gg \omega\varepsilon$ (the $j\omega\varepsilon$ term is negligible)

$\sqrt{j} = e^{j45^\circ} = e^{j\frac{\pi}{4}}$

$$\overline{\eta} = \frac{1+j}{\sqrt{2}}\sqrt{\frac{\omega\mu}{\sigma}} = (1+j)\sqrt{\frac{\omega\mu}{2\sigma}} \rightarrow |\eta| e^{j\tau}$$

$$\boxed{|\overline{\eta}| = \sqrt{\frac{\omega\mu}{\sigma}}} \qquad \boxed{\tau = \pi/4}$$

E and **H** are 45 deg out of phase

Good Conductor: Skin Depth

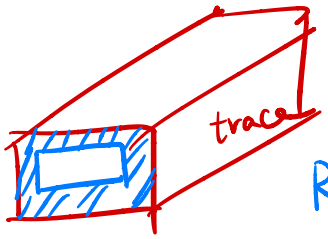
α tells us how far the E and H fields can propagate into a good conductor

Skin Depth δ = Distance that the field strength is attenuated by e^{-1}

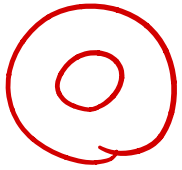
$e^{-\alpha z} = e^{-1}$

$\delta = \frac{1}{\alpha}$

In a conductor, α=LARGE, so fields drop off in a short distance



$R = \frac{L}{\sigma A}$



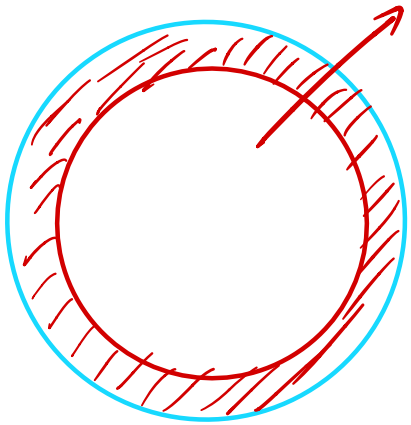
Or

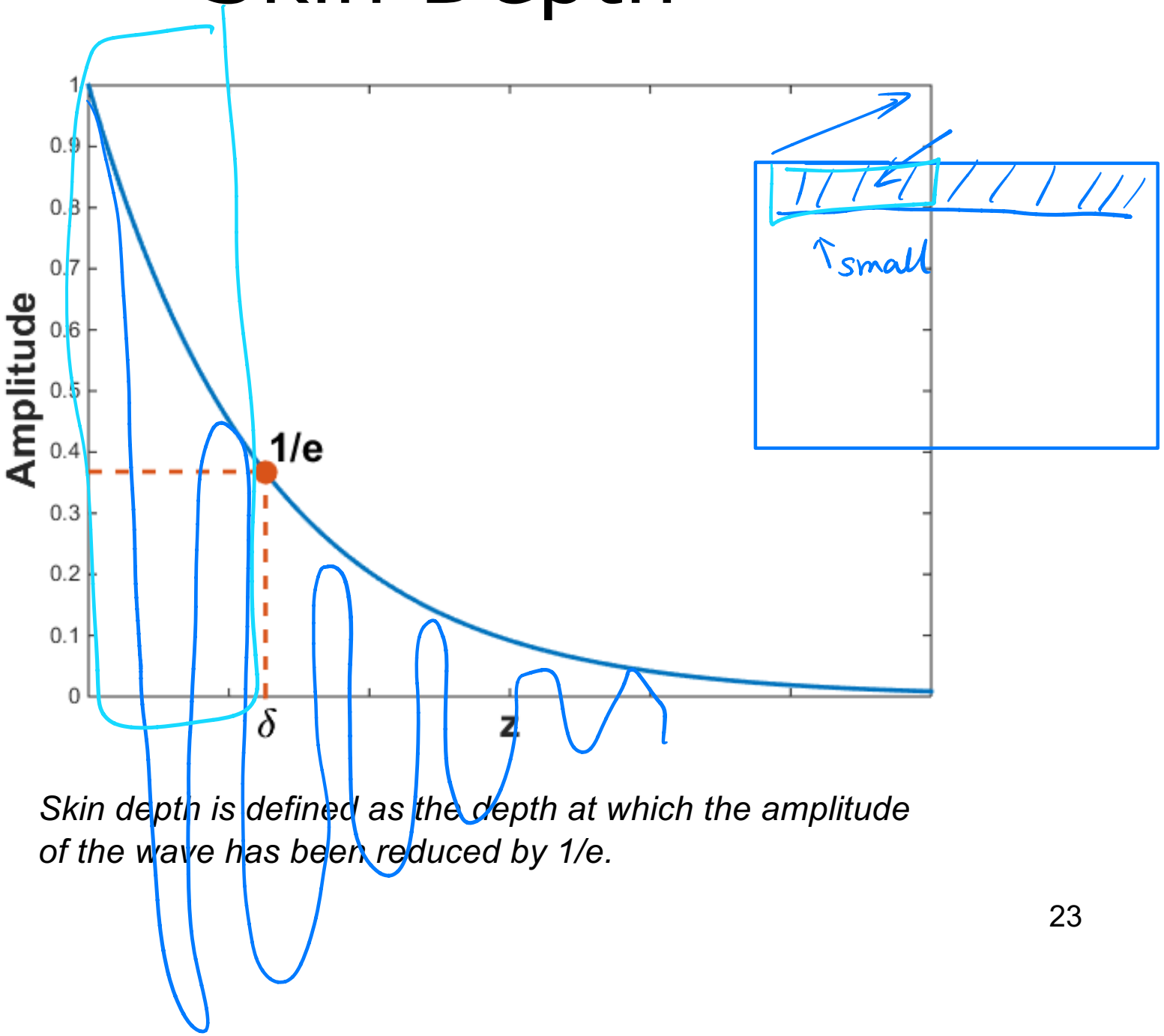
$$\delta = \frac{1}{\alpha} = \sqrt{\frac{2}{\omega\mu\sigma}}$$

$$\delta = \frac{1}{\alpha} = \sqrt{\frac{1}{\pi f\mu\sigma}}$$

Using $\omega = 2\pi f$

Skin Depth





Skin depth is defined as the depth at which the amplitude of the wave has been reduced by 1/e.

Perfect Conductor

$$\sigma = \infty$$

$$\alpha = \omega\sqrt{\frac{\mu\varepsilon}{2}}\left(\sqrt{1+\left(\frac{\sigma}{\omega\varepsilon}\right)^2}-1\right)^{1/2}$$

$$\alpha = \infty$$

$$\delta = 0$$

Infinite attenuation

Means that neither **E** or **H** can propagate into a perfect conductor

NO TIME-VARYING FIELDS (**E** or **H**) CAN EXIST IN A PERFECT CONDUCTOR!!!

Lecture 22b-23 Summary

Perfect Dielectric $\sigma = 0$

Attenuation: $\alpha = 0$

Phase const. $\beta = \omega\sqrt{\epsilon\mu}$

E,**H** In Phase: $\tau = 0$

Impedance: $|\overline{\eta}| = \eta_0\sqrt{\mu_r/\varepsilon_r}$

Speed: $v_p = c/\sqrt{\mu_r\varepsilon_r} \le c$

Wavelength: $\lambda = 2\pi/\beta = \frac{2\pi}{\omega\sqrt{\epsilon\mu}}$

Skin depth: $\delta = 1/\alpha = \infty$

Imperfect Dielectric $\sigma/\omega\varepsilon << 1$

Attenuation: $\alpha = \frac{\sigma}{2}\sqrt{\frac{\mu}{\epsilon}}$

Phase const. $\beta \approx \omega\sqrt{\epsilon\mu}$

E,**H** In Phase: $\tau \approx \sigma/2\omega\epsilon \to 0$

Impedance: $|\overline{\eta}| \approx \eta_0\sqrt{\mu_r/\varepsilon_r}$

Speed: $v_p \approx c/\sqrt{\mu_r\varepsilon_r} \le c$

Wavelength: $\lambda = \frac{2\pi}{\omega\sqrt{\epsilon\mu}}$

Skin depth: $\delta \approx \frac{2}{\sigma}\sqrt{\frac{\epsilon}{\mu}}$

Good Conductor $\sigma/\omega\varepsilon >> 1$

Attenuation: $\alpha \approx \sqrt{\omega\mu\sigma/2} = \sqrt{\pi f\mu\sigma}$

Phase const. $\beta \approx \sqrt{\omega\mu\sigma/2} = \sqrt{\pi f\mu\sigma}$

E,**H** 45° Phase: $\tau \approx 45°$

Impedance: $|\overline{\eta}| \approx \sqrt{\omega\mu/\sigma}$

Speed: $v_p \approx \sqrt{2\omega/\sigma\mu}$

Wavelength: $\lambda \approx 2\pi/\sqrt{\pi f\mu\sigma}$

Skin depth: $\delta = \frac{1}{\alpha} \approx \frac{1}{\sqrt{\pi f\mu\sigma}}$

Perfect Conductor $\sigma \to \infty$

Attenuation: $\alpha \to \infty$

Phase const. $\beta \to \infty$

E,**H** 45° Phase: $\tau \to \pi/4$

Impedance: $|\overline{\eta}| \to 0$

Speed: $v_p \to 0$

Wavelength: $\lambda \to 0$

Skin depth: $\delta \to 0$

$\vec{E} \to 0$

$\vec{H} \to 0$